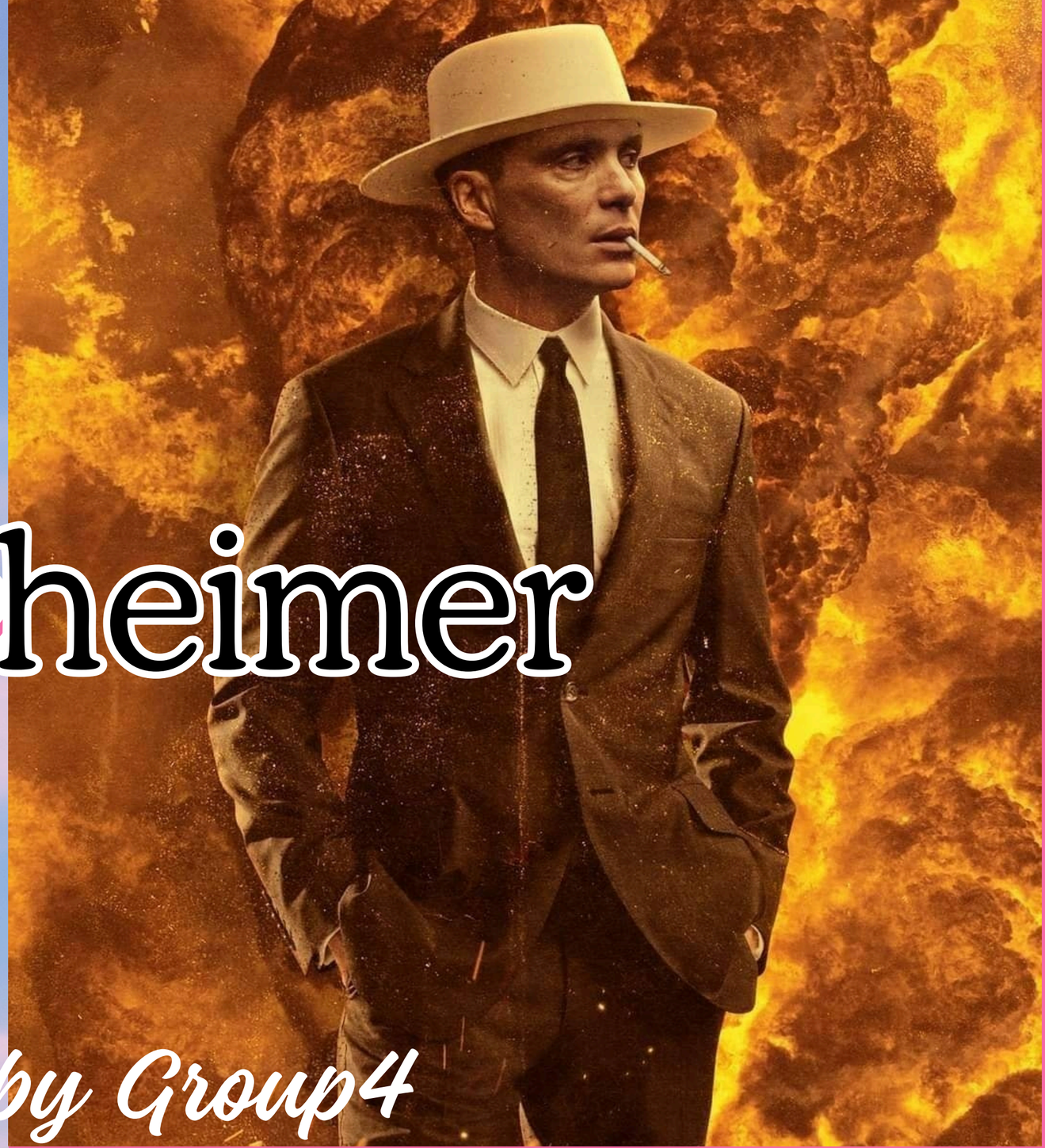




Barbenheimer

Presented by Group4

Diving into the *Barben*heimer phenomenon

What linguistic shifts occurred in audience discussions of Barbie and Oppenheimer from pre-release speculation to post-release reception?

Barbenheimer

Data Collection: Pink Meets the Atomic Age

How data was gathered, sources used, and methodology

1

Kaggle Dataset 1

- Youtube Comments of from Barbie (2023) Trailer, Oppenheimer Trailer, and Pentex Productions video where two movies were compared (Barbenheimer)
- x3 CSV Files (.csv) imported into Google Colab

2

Kaggle Dataset 2

- IMDB reviews for Barbie and Oppenheimer
- x 2 CSV Files (.csv) imported into Google Colab

Barbenheimer

Pre-Processing: Cleaning the Glitter & Fallout

1. Removal of non-textual elements
2. Slang conversion to full words
3. Tokenization
4. Lemmatizing
5. Lowercasing
6. Removal of Stop words, punctuations, meaningless words and digits
7. POS (part of speech) tagging

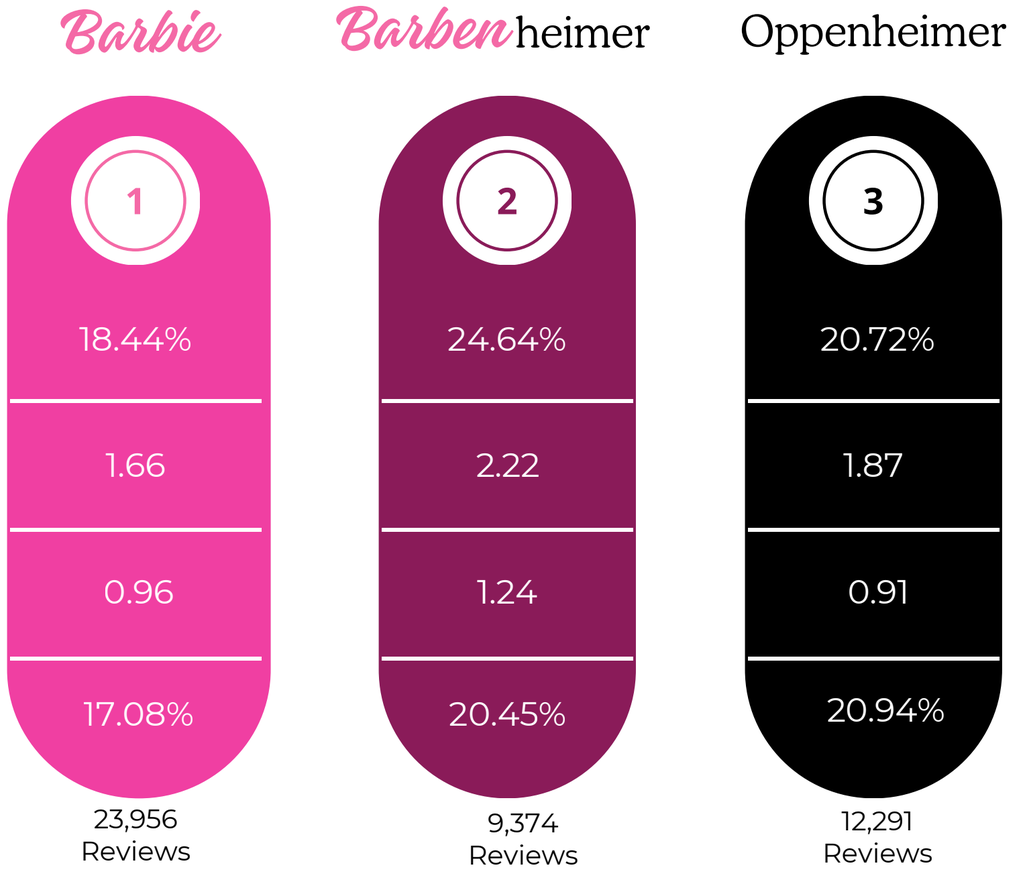
Barbenheimer

Sentiment Analysis: The Mood of the Moviegoers

PRE-RELEASE

The Lexical Suite

- Overall Sentiment
- Avg. Valence Score
  Positivity-Negativity Intensity
- Avg. Extremity Score
  Degree of Positivity or Negativity
- Ambivalence



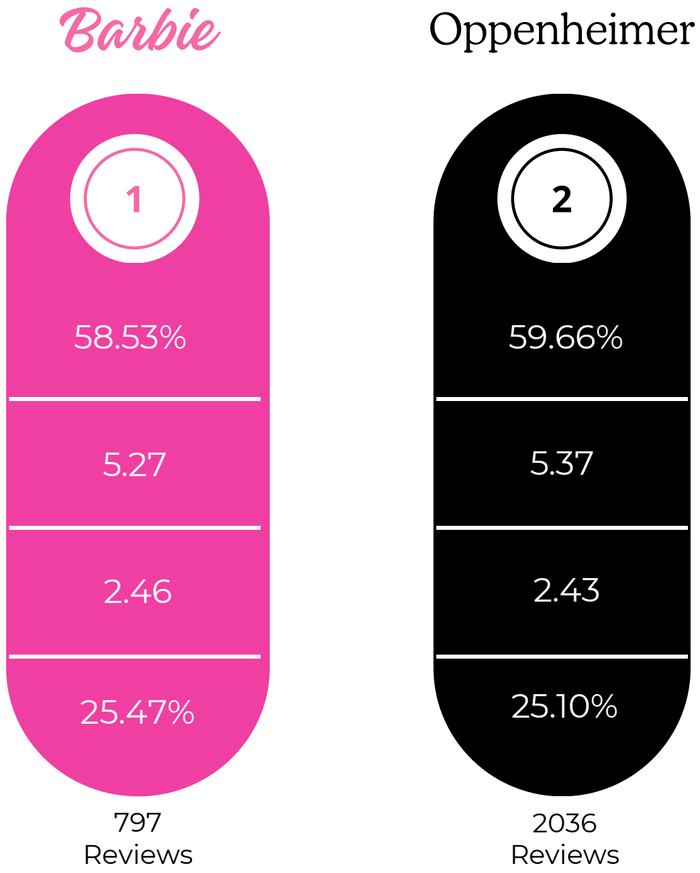
Barbenheimer

Sentiment Analysis: The Mood of the Moviegoers

POST-RELEASE

The Lexical Suite

- Overall Sentiment
- Avg. Valence Score
  Positivity-Negativity Intensity
- Avg. Extremity Score
  Degree of Positivity or Negativity
- Ambivalence



Barbenheimer

Topic Modelling: What Are People Talking About?

Visualizing key discussions surrounding Barbenheimer

Topic Modelling (Latent Dirichlet Allocation)

```
🔷 Topics for Barbie (Pre-Release YouTube):

(0, '0.068*"br" + 0.043*"movie" + 0.022*"barbie" + 0.011*"
(1, '0.019*"``" + 0.018*"movie" + 0.017*"href=" + 0.016*"ba
(2, '0.044*"movie" + 0.043*"barbie" + 0.028*"watch" + 0.013

🔷 Topics for Oppenheimer (Pre-Release YouTube):

(0, '0.064*"br" + 0.028*"movie" + 0.022*"quot" + 0.013*"wat
(1, '0.020*"movie" + 0.014*"oppenheimer" + 0.013*"film" + 0
(2, '0.018*"movie" + 0.014*"see" + 0.008*"bomb" + 0.008*"wa

🔷 Topics for Barbie (Post-Release IMDB):

(0, '0.028*"barbie" + 0.027*"movie" + 0.015*"helpful" + 0.0
(1, '0.039*"helpful" + 0.021*"find" + 0.020*"review" + 0.02
(2, '0.023*"barbie" + 0.015*"film" + 0.015*"movie" + 0.010*

🔷 Topics for Oppenheimer (Post-Release IMDB):

(0, '0.024*"movie" + 0.017*"oppenheimer" + 0.014*"good" + 0
(1, '0.021*"movie" + 0.018*"oppenheimer" + 0.013*"nolan" +
(2, '0.023*"oppenheimer" + 0.017*"be" + 0.013*"nolan" + 0.0
/usr/local/lib/python3.11/dist-packages/ipykernel/ipkernel.
  and should_run_async(code)
```

Key insights from Topic Modelling

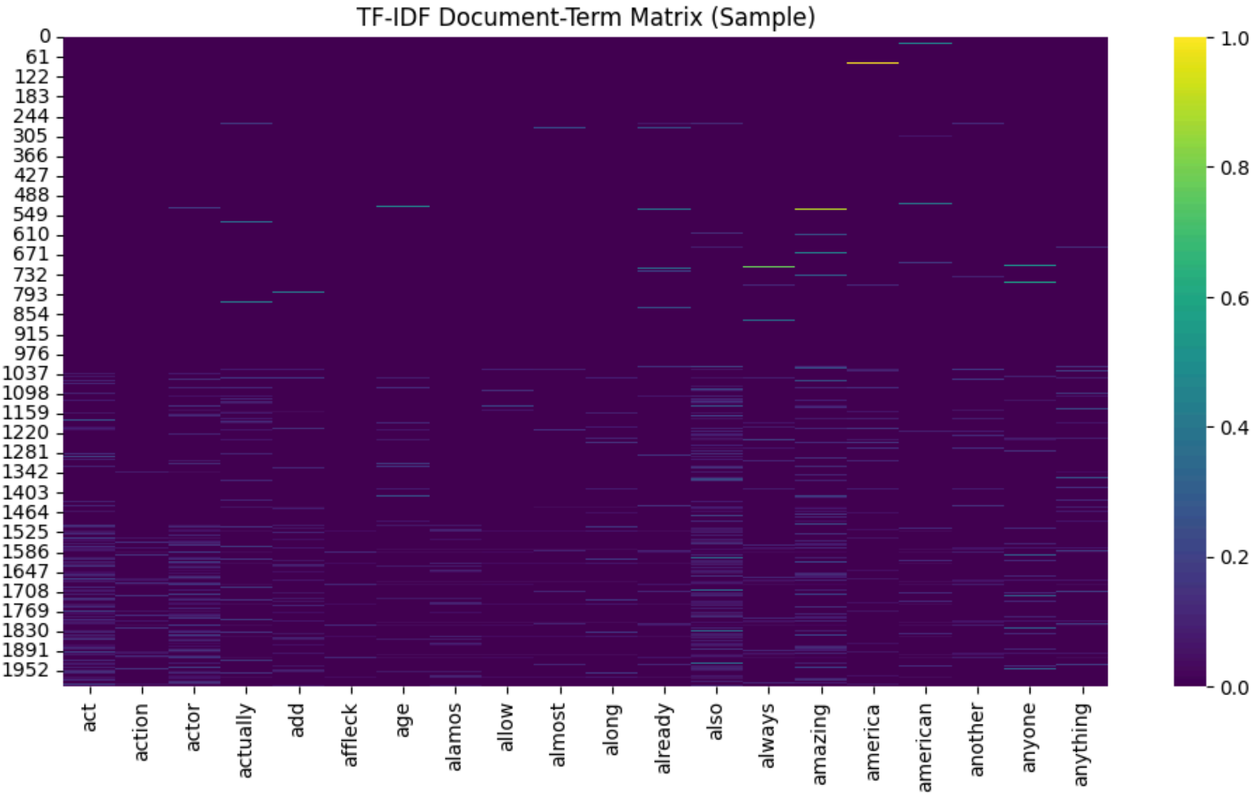
1. **Pre-release**: Focus on anticipation, marketing, and comparisions
2. **Post-release:** Discussions shift to reviews, awards, and key scenes.
3. **Barbie vs. Oppenheimer:** Fun & culture vs. history & war themes.
4. **IMDb Impact**: User ratings & spoilers shape film reception.

Barbenheimer

From Pink to Atomic: TF-IDF Vectorization of Barbenheimer

Visualizing key discussions surrounding Barbenheimer

Vectorization (TF-IDF)

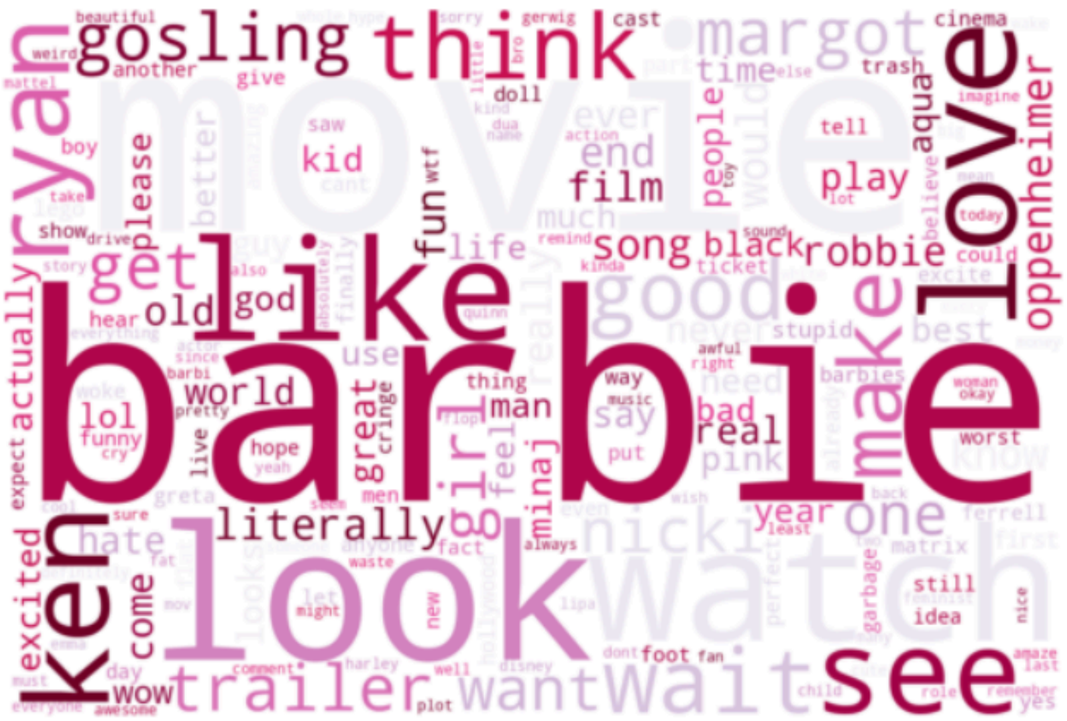


Key insights from Vectorization of Barbenheimer

1. **Key Terms:** "Barbie" & "Oppenheimer" have high TF-IDF, defining their reviews.
2. **Common Words:** Generic terms like "amazing" & "bad" appear often but may have low TF-IDF.
3. **Sentiment Words:** Terms like "worth" & "young" reflect user perspectives. .
4. **Review Clusters:** Heatmap shows word groupings, indicating shared opinions.

Word Clouds: What Are People Talking About?

Visualizing key discussions surrounding Barbenheimer



Barbie

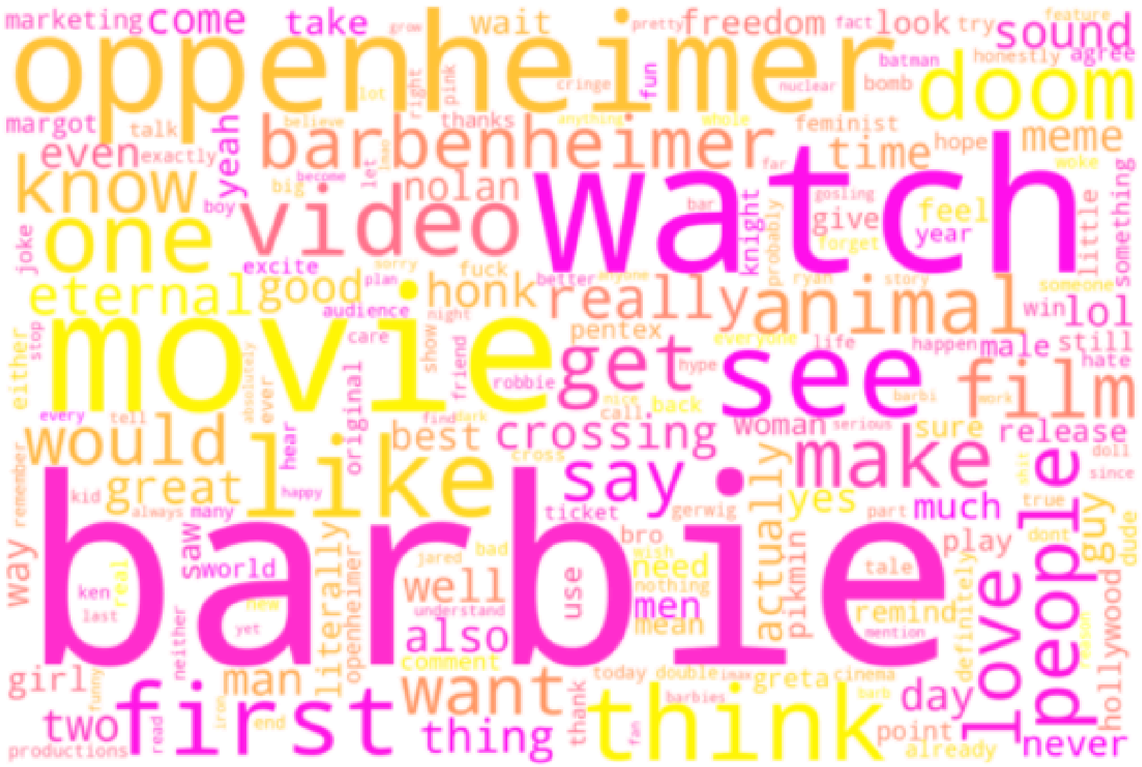


Oppenheimer

Barbenheimer

Word Clouds: What Are People Talking About?

Visualizing key discussions surrounding Barbenheimer



- Strong Viewer Engagement
  "Barbie", "Oppenheimer", “Make”
- Positive Emotions
  "Like", "Great"
- Virality and Word-of-Mouth
  "Go", "Want", “Watch”
- Barbie - Eliciting Strong Positive Emotions
  "Love"
- Oppenheimer - Serious Themes
  "Doom"
- Comparative Insights
  "First", “Two”, “Time”

Barbenheimer

Conclusion



Our analysis shows a clear shift in audience discussions: pre-release focused on hype and comparisons, while post-release emphasized storytelling, performances, and themes.

Pre-release discussions were speculative, while post-release became analytical, reflecting the shift from anticipation to critical assessment and the impact of cinematic framing on public discourse.

Thank you

